Correlation

The correlation between two stocks ρ_{ij} is the covariance of the stocks divided by the volatility of each stock. This provides a correlation coefficient between one and minus one: $-1 \leq \rho_{ij} \leq 1$.

$$\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \cdot \sigma_j} \tag{6.8}$$

Stocks with a correlation of $\rho_{ij} = 1$ move perfectly with one another, stocks with a correlation of $\rho_{ij} = -1$ move perfectly in the opposite direction of one another, and stocks with a correlation of $\rho_{ij} = 0$ do not move together at all. Correlation provides a measure of the strength of co-movement between stocks.

Dispersion

The dispersion of returns is computed as the standard deviation of returns for a group of stocks. It is a cross-sectional measure of overall variability across stocks.

$$dispersion(r_p) = \sqrt{\frac{1}{n-1} \cdot \sum_{j=1}^{n} (\bar{r}_j - \bar{r}_p)^2} \tag{6.9}$$

where $\bar{r}_j$ is the average return for the stock and $\bar{r}_p$ is the average return across all stocks in the sample.

Dispersion is very useful to portfolio managers because it gives a measure of the directional movement of prices and how close they are moving in conjunction with one another. A small dispersion measure indicates that the stocks are moving up and down together. A large dispersion measure indicates that the stocks are not moving closely together.

Value-at-Risk

Value-at-risk (VaR) is a summary statistic that quantifies the potential loss of a portfolio. Many companies place limits on the total value-at-risk to protect investors from potential large losses. This potential loss corresponds to a specified probability α level or alternatively a $(1-\alpha)$ confidence.

If the expected return profile for a portfolio is $r \sim N(\bar{r}_p, \sigma_p^2)$. Then a $\alpha\%$ VaR estimate is the value return that occurs at the $1-\alpha$ probability level in the cumulative normal distribution. If $\alpha = 95\%$ this equation is:

$$0.05 = \int_{-\infty}^{r*} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(r-\bar{r}_p)^2}{2\sigma_p} \right\} \tag{6.10}$$